

Experiment 2: Output

Automobile_2a:

	symboling	normalized-losses	make	fuel-type	...	peak-rpm	city-mpg	highway-mpg	price
0	3	NaN	alfa-romero	gas	...	5000.0	21	27	13495.0
1	3	NaN	alfa-romero	gas	...	5000.0	21	27	16500.0
2	1	NaN	alfa-romero	gas	...	5000.0	19	26	16500.0
3	2	164.0	audi	gas	...	5500.0	24	30	13950.0
4	2	164.0	audi	gas	...	5500.0	18	22	17450.0

[5 rows x 26 columns]

	symboling	normalized-losses	make	fuel-type	...	peak-rpm	city-mpg	highway-mpg	price
0	3	NaN	alfa-romero	gas	...	5000.0	21	27	13495.0
1	3	NaN	alfa-romero	gas	...	5000.0	21	27	16500.0
2	1	NaN	alfa-romero	gas	...	5000.0	19	26	16500.0
3	2	164.0	audi	gas	...	5500.0	24	30	13950.0
4	2	164.0	audi	gas	...	5500.0	18	22	17450.0
..	...	...	...	...	...	...	...	...	...
200	-1	95.0	volvo	gas	...	5400.0	23	28	16845.0
201	-1	95.0	volvo	gas	...	5300.0	19	25	19045.0
202	-1	95.0	volvo	gas	...	5500.0	18	23	21485.0
203	-1	95.0	volvo	diesel	...	4800.0	26	27	22470.0
204	-1	95.0	volvo	gas	...	5400.0	19	25	22625.0

[205 rows x 26 columns]

Data Types

symboling	int64
normalized-losses	float64
make	str
fuel-type	str
aspiration	str
num-of-doors	str
body-style	str
drive-wheels	str
engine-location	str
wheel-base	float64
length	float64
width	float64
height	float64
curb-weight	int64
engine-type	str

```
num-of-cylinders    str
engine-size         int64
fuel-system         str
bore                float64
stroke              float64
compression-ratio   float64
horsepower          float64
peak-rpm            float64
city-mpg            int64
highway-mpg         int64
price               float64
dtype: object
```

Dataset Information

<class 'pandas.DataFrame'>

RangeIndex: 205 entries, 0 to 204

Data columns (total 26 columns):

#	Column	Non-Null Count	Dtype
0	symboling	205 non-null	int64
1	normalized-losses	164 non-null	float64
2	make	205 non-null	str
3	fuel-type	205 non-null	str
4	aspiration	205 non-null	str
5	num-of-doors	203 non-null	str
6	body-style	205 non-null	str
7	drive-wheels	205 non-null	str
8	engine-location	205 non-null	str
9	wheel-base	205 non-null	float64
10	length	205 non-null	float64
11	width	205 non-null	float64
12	height	205 non-null	float64
13	curb-weight	205 non-null	int64
14	engine-type	205 non-null	str
15	num-of-cylinders	205 non-null	str
16	engine-size	205 non-null	int64
17	fuel-system	205 non-null	str
18	bore	201 non-null	float64
19	stroke	201 non-null	float64
20	compression-ratio	205 non-null	float64
21	horsepower	203 non-null	float64
22	peak-rpm	203 non-null	float64
23	city-mpg	205 non-null	int64
24	highway-mpg	205 non-null	int64

25 price 201 non-null float64
dtypes: float64(11), int64(5), str(10)
memory usage: 50.2 KB

Shape of Dataset
(205, 26)

Column Names

```
Index(['symboling', 'normalized-losses', 'make', 'fuel-type', 'aspiration',  
      'num-of-doors', 'body-style', 'drive-wheels', 'engine-location',  
      'wheel-base', 'length', 'width', 'height', 'curb-weight', 'engine-type',  
      'num-of-cylinders', 'engine-size', 'fuel-system', 'bore', 'stroke',  
      'compression-ratio', 'horsepower', 'peak-rpm', 'city-mpg',  
      'highway-mpg', 'price'],  
      dtype='str')
```

Statistical Summary

	symboling	normalized-losses	wheel-base	...	city-mpg	highway-mpg	price
count	205.000000	164.000000	205.000000	...	205.000000	205.000000	201.000000
mean	0.834146	122.000000	98.756585	...	25.219512	30.751220	13207.129353
std	1.245307	35.442168	6.021776	...	6.542142	6.886443	7947.066342
min	-2.000000	65.000000	86.600000	...	13.000000	16.000000	5118.000000
25%	0.000000	94.000000	94.500000	...	19.000000	25.000000	7775.000000
50%	1.000000	115.000000	97.000000	...	24.000000	30.000000	10295.000000
75%	2.000000	150.000000	102.400000	...	30.000000	34.000000	16500.000000
max	3.000000	256.000000	120.900000	...	49.000000	54.000000	45400.000000

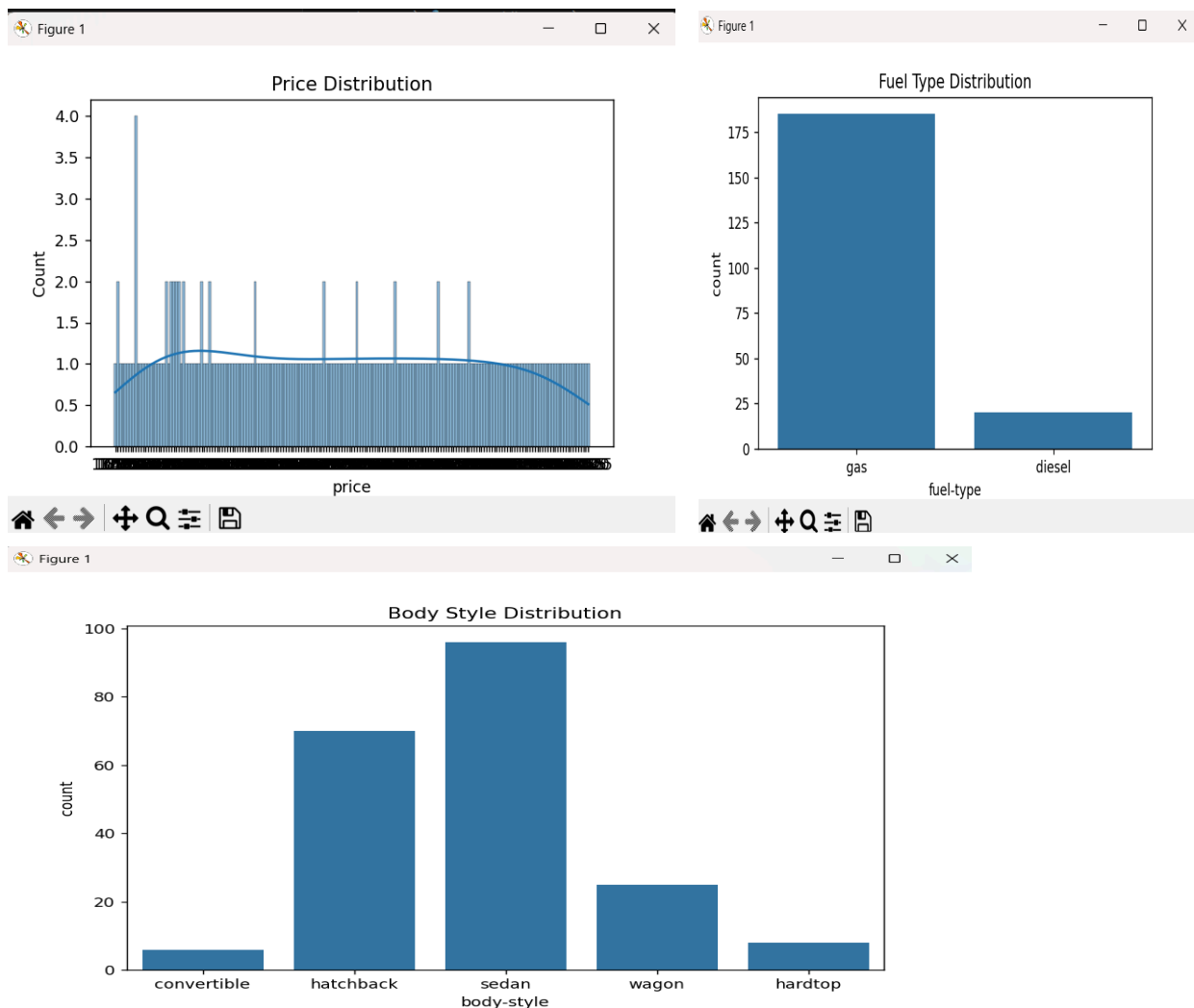
[8 rows x 16 columns]

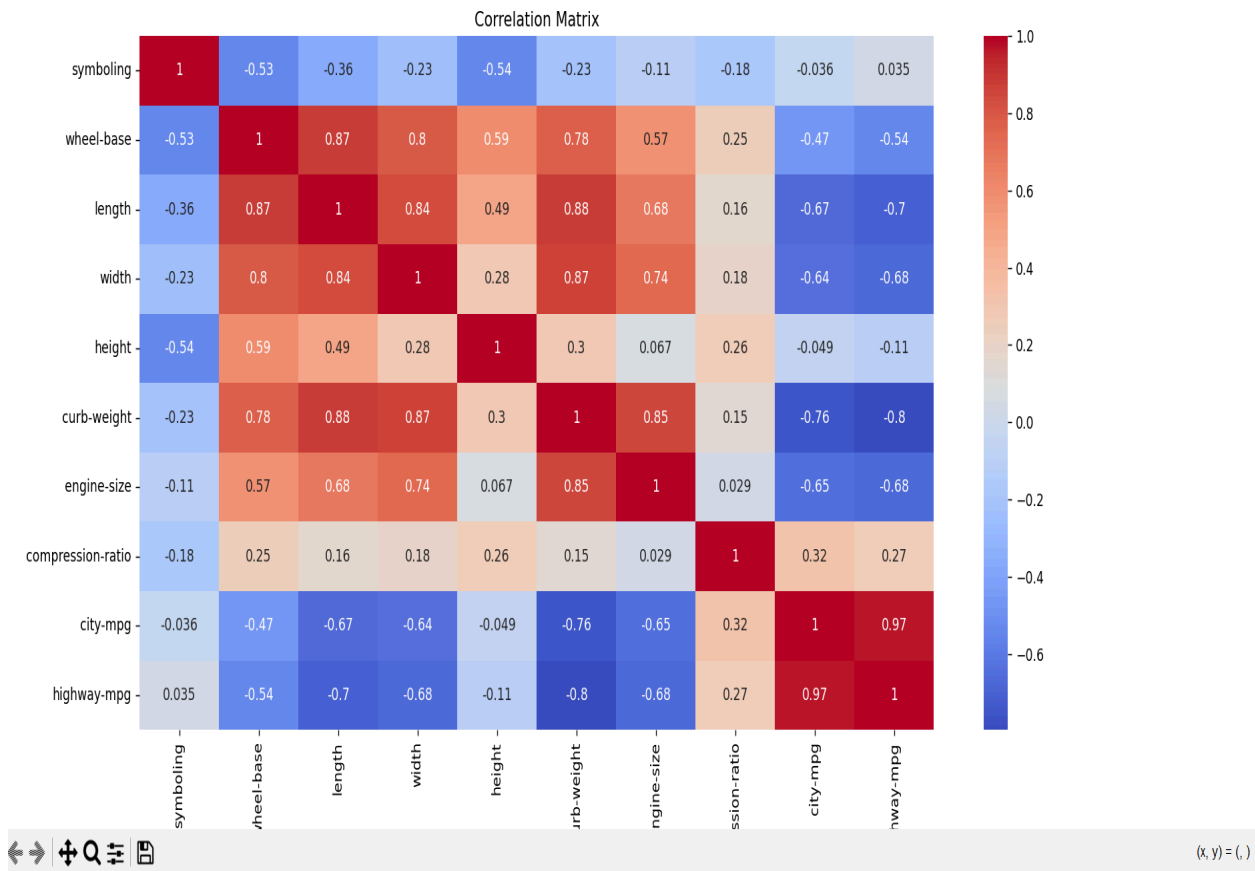
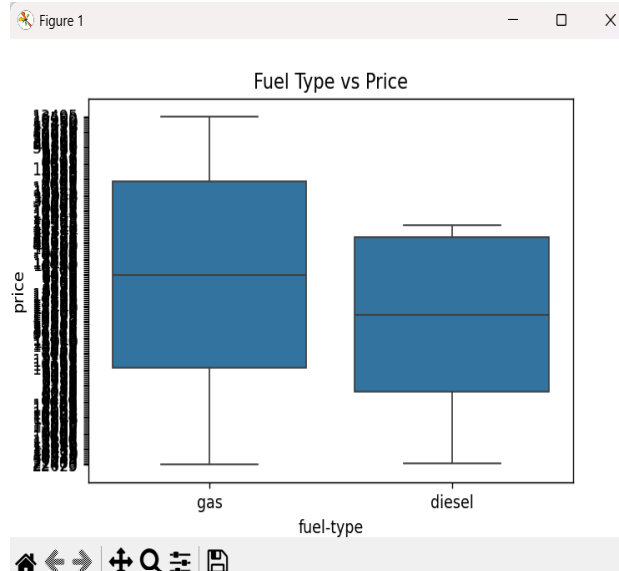
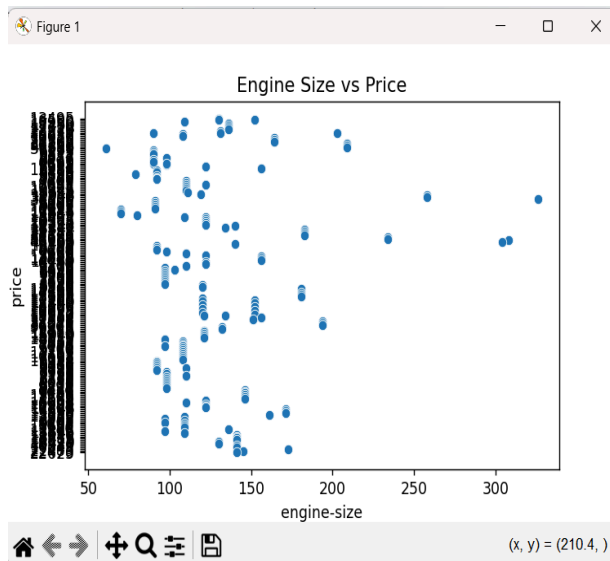
Missing Values

symboling	0
normalized-losses	41
make	0
fuel-type	0
aspiration	0
num-of-doors	2
body-style	0
drive-wheels	0
engine-location	0
wheel-base	0
length	0
width	0
height	0
curb-weight	0

```
engine-type      0
num-of-cylinders 0
engine-size      0
fuel-system      0
bore             4
stroke           4
compression-ratio 0
horsepower       2
peak-rpm         2
city-mpg         0
highway-mpg      0
price            4
dtype: int64
```

Automobile_2a_part2:





exp-2_Automobile_2b:

Numeric Attributes

```
Index(['symboling', 'wheel-base', 'length', 'width', 'height', 'curb-weight',  
      'engine-size', 'compression-ratio', 'city-mpg', 'highway-mpg'],  
      dtype='str')
```

Nominal Attributes

```
['make', 'fuel-type', 'aspiration', 'body-style', 'drive-wheels', 'engine-location', 'engine-type',  
'num-of-cylinders', 'fuel-system']
```

Binary Attributes

```
['fuel-type', 'aspiration', 'engine-location']
```

exp-2_Automobile_2c:

Nominal Attribute Dissimilarity

Fuel Type vs Aspiration = 1.0